



The Biography of the Search Engine
That Organized the World

BASED ON THE BOOK BY STEVEN LEVY

The Information Prosthesis

In 2007, a Google delegation visited Ragihalli, a rural Indian village. A villager there had never heard the word “Google,” yet he held a cell phone with with four bars of signal.



*“This man might not know Google yet... And then he **would** know Google.”*

Creative Disorganization



The Scene:

A landscape of exercise balls and unfilled cubicles.

The Culture:

Run by engineers, for engineers. Obsessed with data, disregarding corporate norms.

The Leadership:

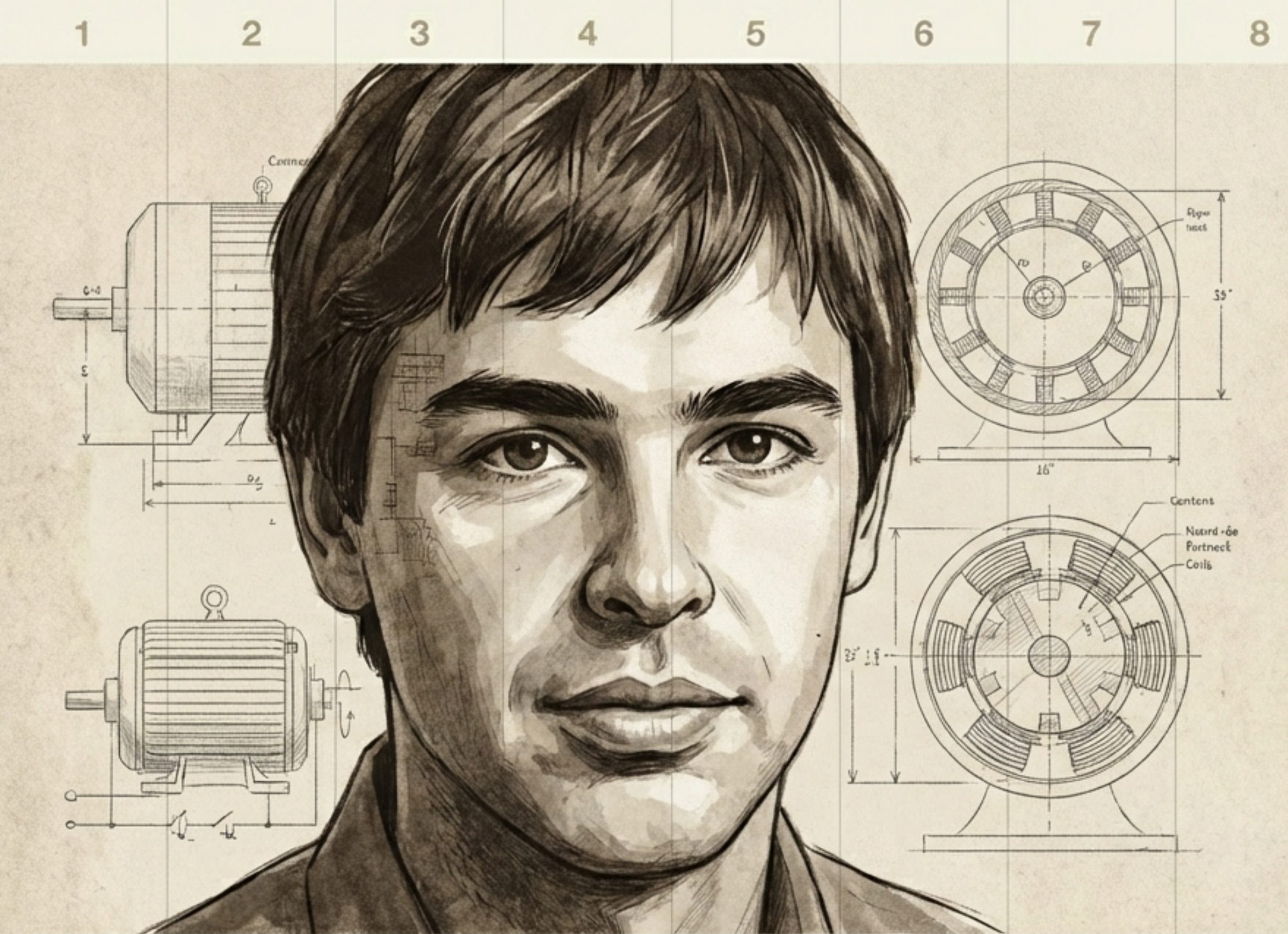
“LarryAndSergey”—a two-headed organism valuing intellect over hierarchy.

Lab Note:

Date: October 1999

Location: The Googleplex

Status: Genial Chaos



Profile: Roboto Mono
Son of computer science professors. The first child in his elementary school to submit a word-processed document.

The Motivation: Roboto Mono
Haunted by Nikola Tesla, the genius inventor who died in poverty.

The Philosophy: Roboto Mono
“I didn’t want to just invent things... you need to do more than just invent things.” Invention without commercialization is failure.”

“A healthy disregard for the impossible.”

The Prodigy: Sergey Brin

Profile:

Russian immigrant. Math prodigy. Entered Stanford PhD program at age 19.

Personality:

Bold, quirky, and athletic.
Traversing Stanford hallways
on Rollerblades.

Academic Style:

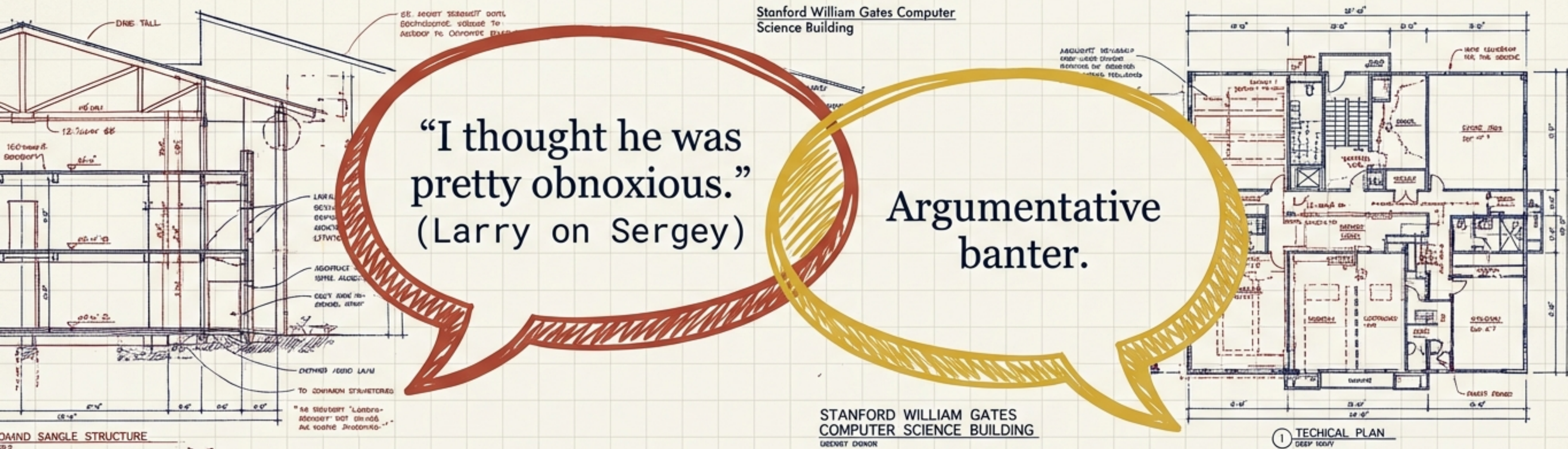
Breezed through required tests to "sample" courses. When pressed on taking advanced classes, he joked he might take "advanced swimming."

Role:

The intellectual tuning fork to Page.



The Collision at Gates Hall

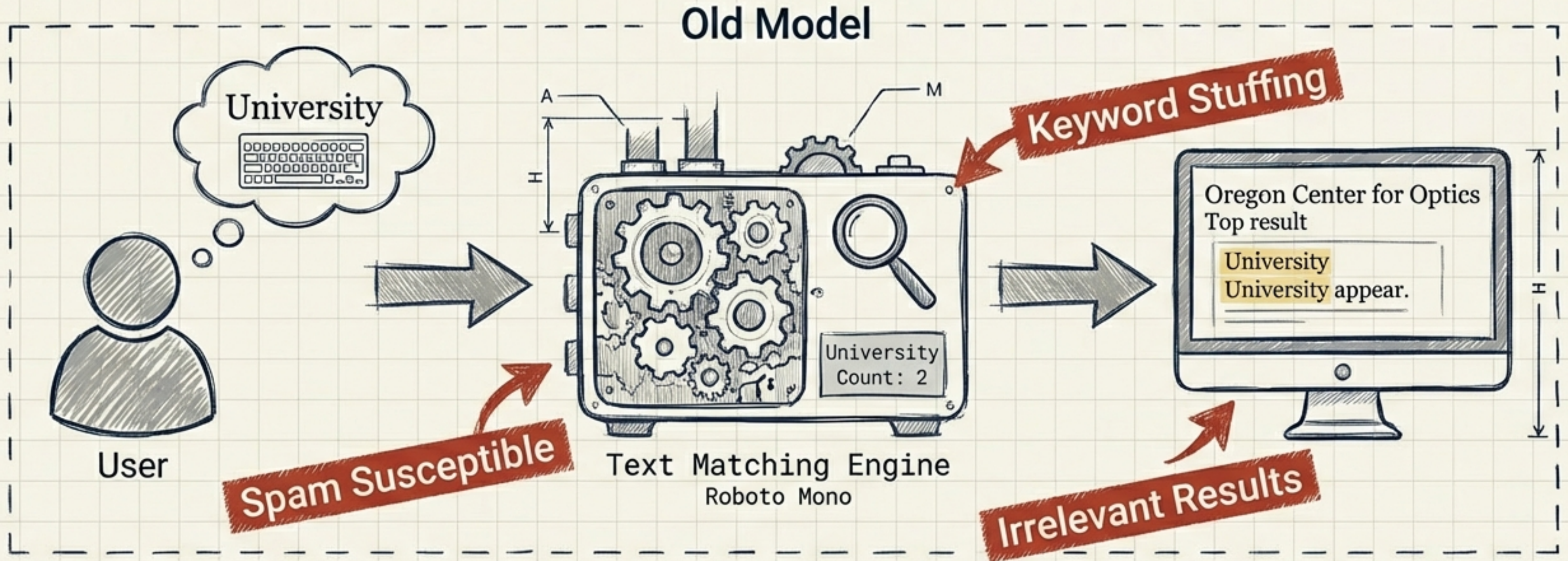


Context: Summer 1995. Sergey Brin is the tour guide; Larry Page is the tourist.

The Dynamic: “LarryAndSergey” was born from intellectual sparring.

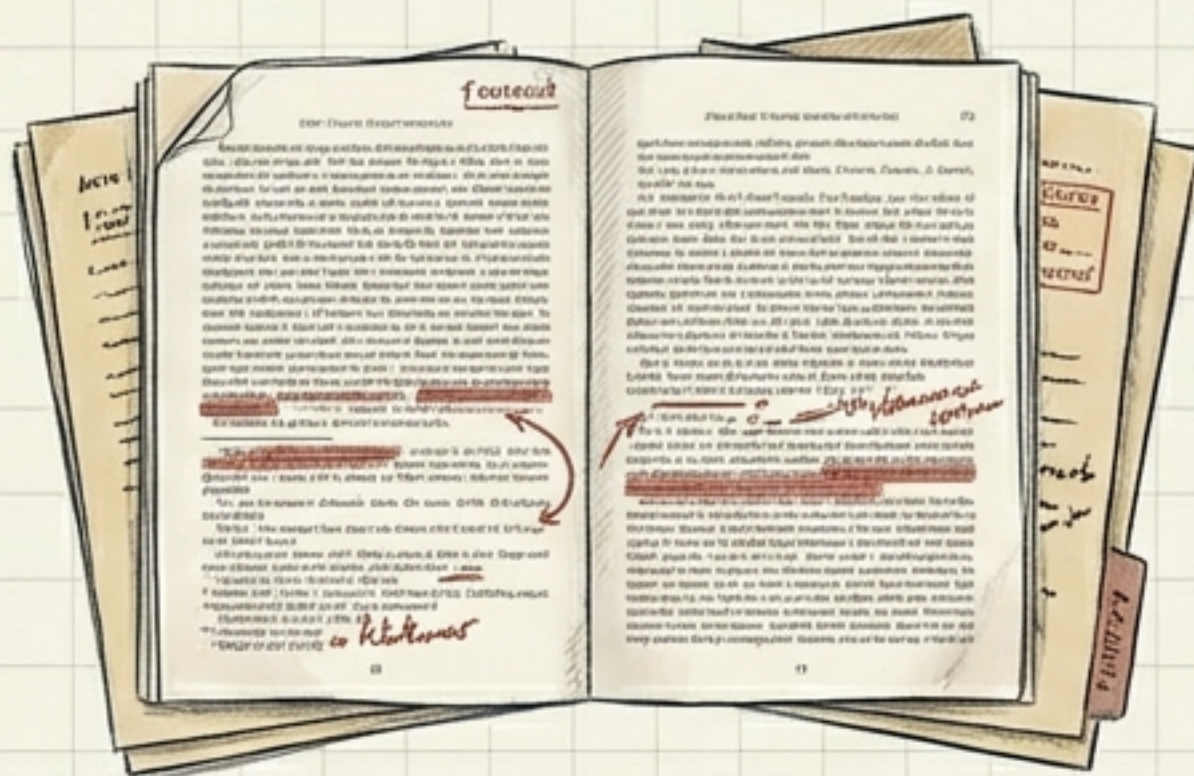
Common Ground: A rock-stubborn belief in the meritocracy of academia and the primacy of data.

The Flaw in the Web (1996).

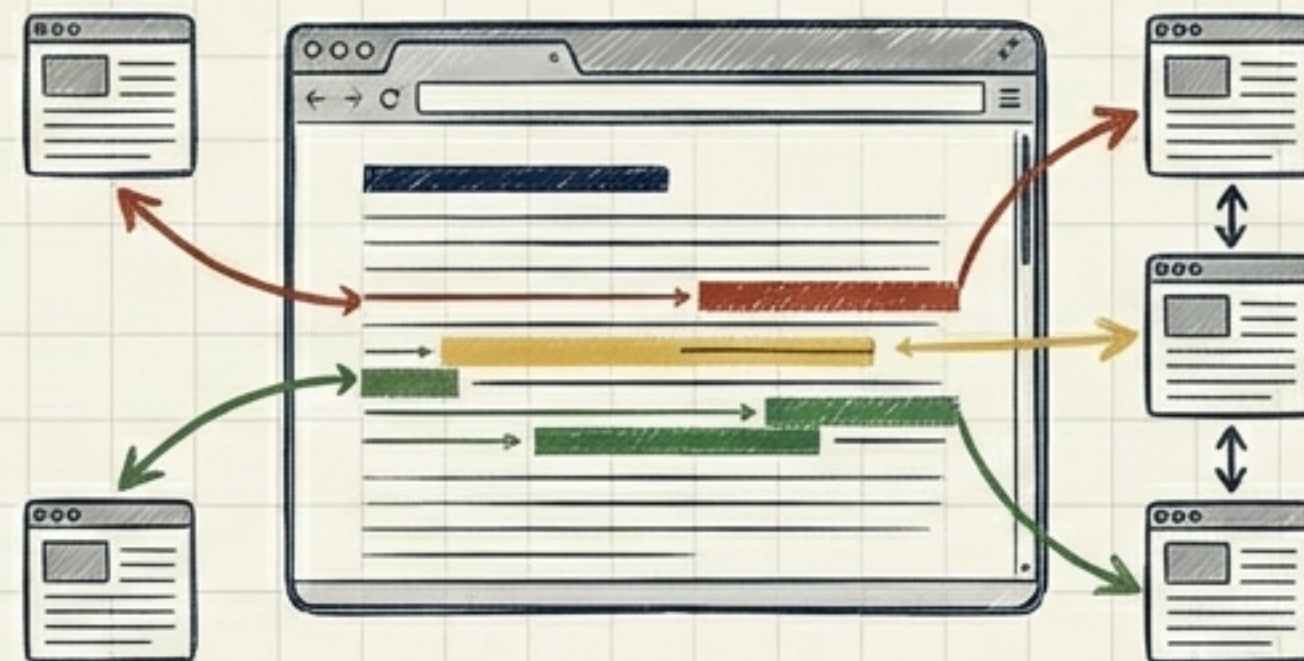


The Industry Attitude: An AltaVista engineer admitted, “Anyone who tells you it’s scientific is just pulling your leg.” Search was considered voodoo.

The Web as a Scholarly Paper.



Citation = Authority.



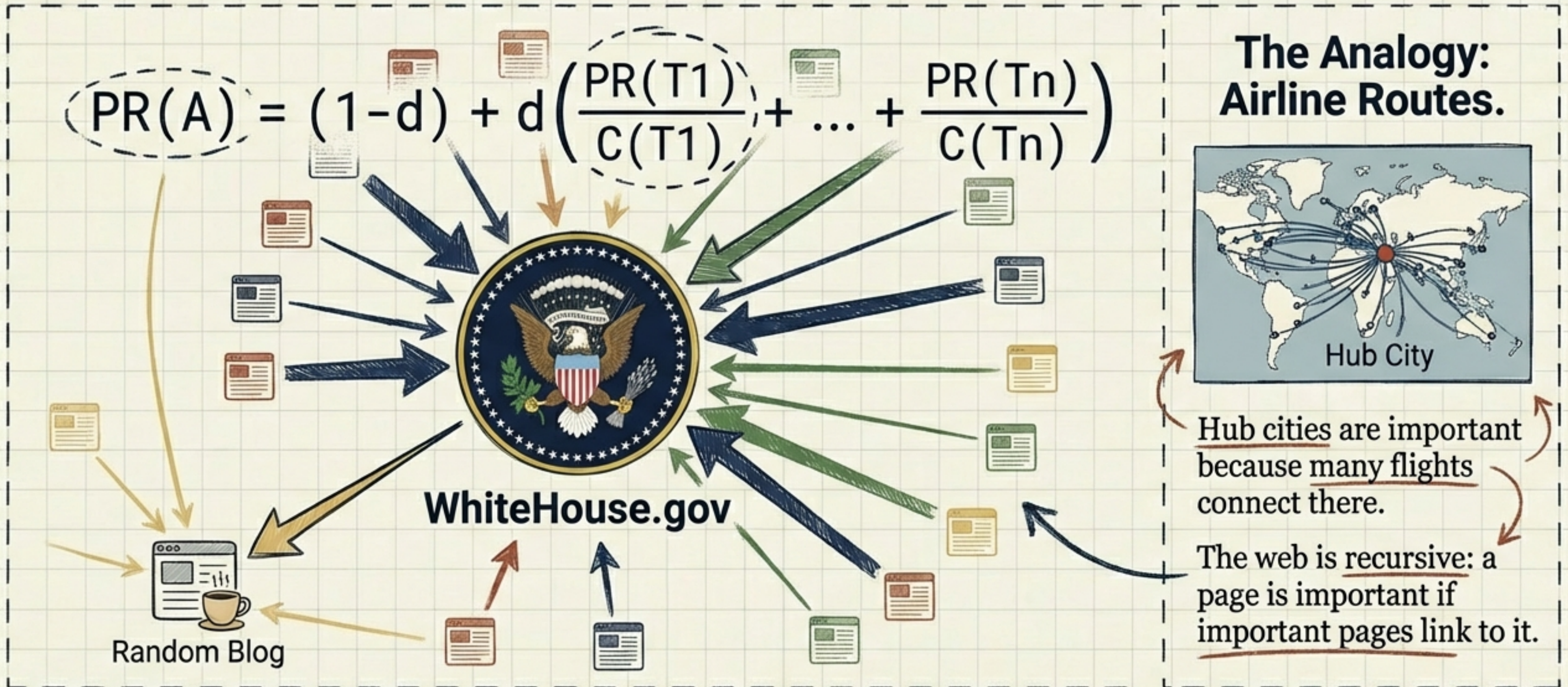
Link = Vote.

The Insight: Larry Page realized the web is a citation graph. If Page A links to Page B, it is a vote of confidence for Page B.

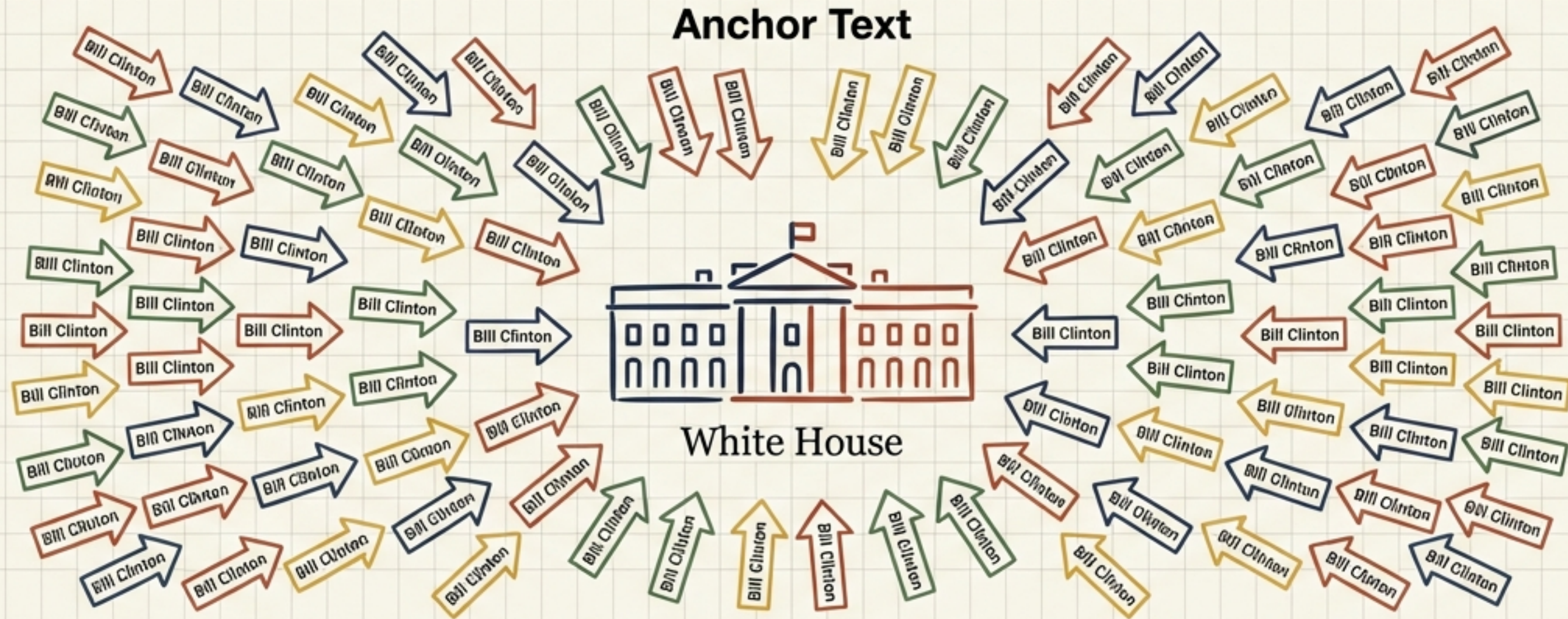
The Project: “BackRub”—crawling the web backwards to see who links to whom.

“The early versions of hypertext had a tragic flaw: you couldn’t follow links in the other direction... BackRub was about reversing that.”

The Math: PageRank.



The 'Bill Clinton' Test



The Innovation: Google analyzed the words *in the link* pointing to the page, not just the page itself

Competitor Result
"Bill Clinton Joke of the Day"
(High text match)

Google Result
"whitehouse.gov"
(High anchor text relevance)

Infrastructure on a Shoestring

The Rig:

Custom-built with Legos for modularity and heat dissipation

The Storage:

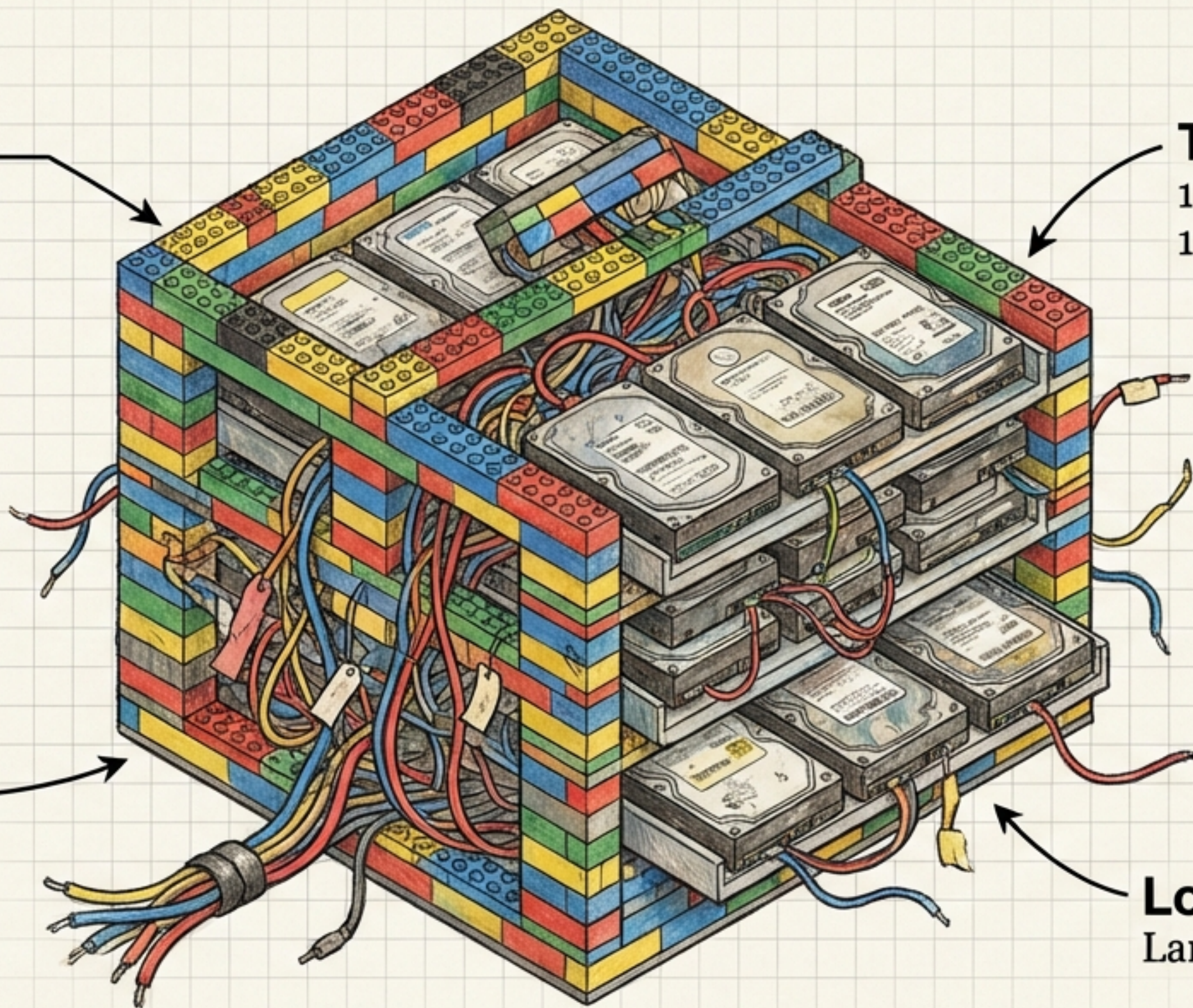
120 scrapped drives =
1 Terabyte

The Bandwidth:

Hacked into the Stanford T3 line, consuming half the university's capacity.

Location:

Larry Page's dorm room.



The Reluctant Entrepreneurs

The Goal: They wanted to be academics, not CEOs. They tried to sell the tech to license it and return to their PhDs.



The Fuki Sushi Meeting.

The Query: "Internet"

Excite (The Portal Model).

Result: Chinese pages with English characters.

Strategy: Stickiness (Keep users reading ads).

Google (The BackRub Model).

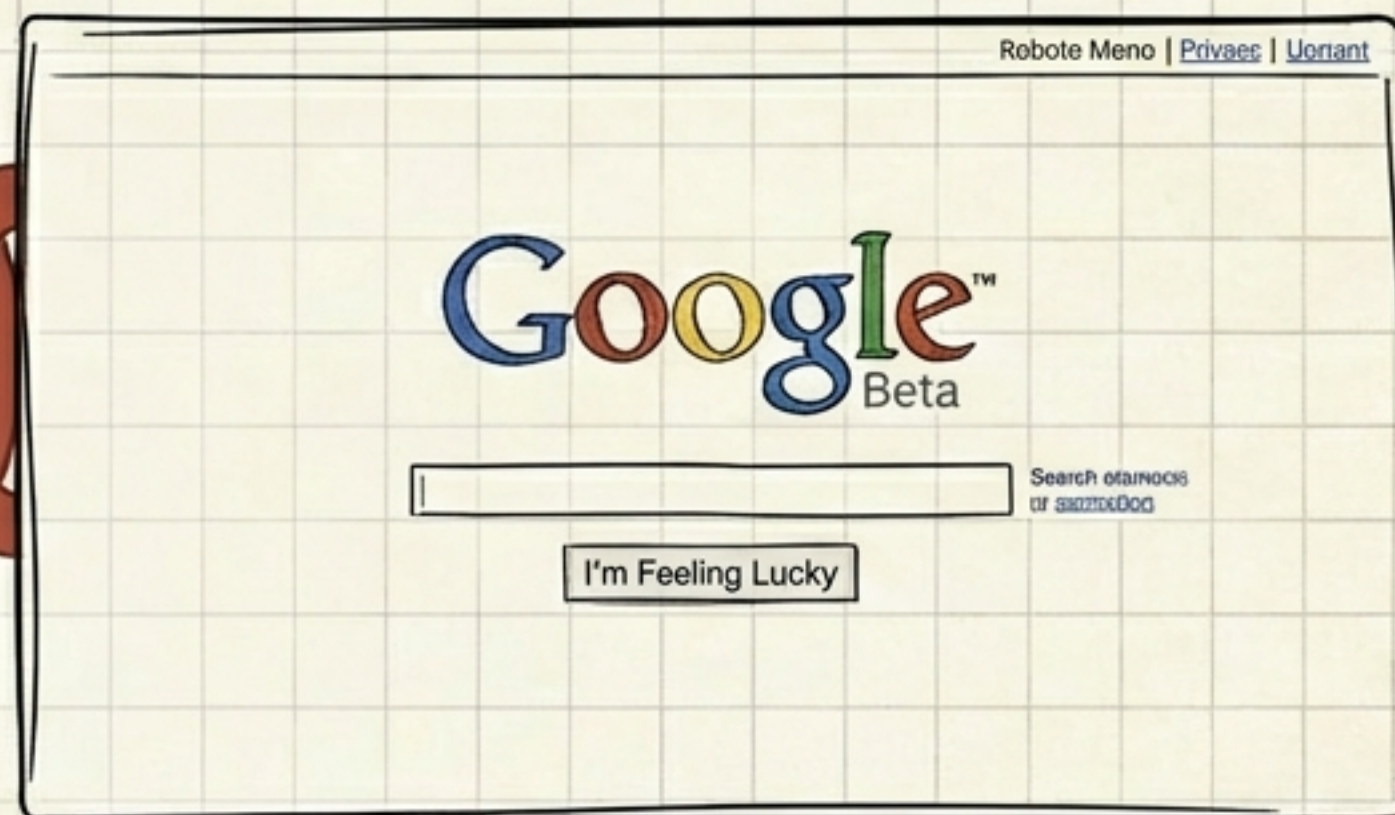
Result: Instructions on using a browser.

Strategy: Speed (Send users away instantly).

George Bell (Excite CEO) wanted a search engine only "80% as good" so users wouldn't leave too fast. The deal died.

1 From BackRub to Google.

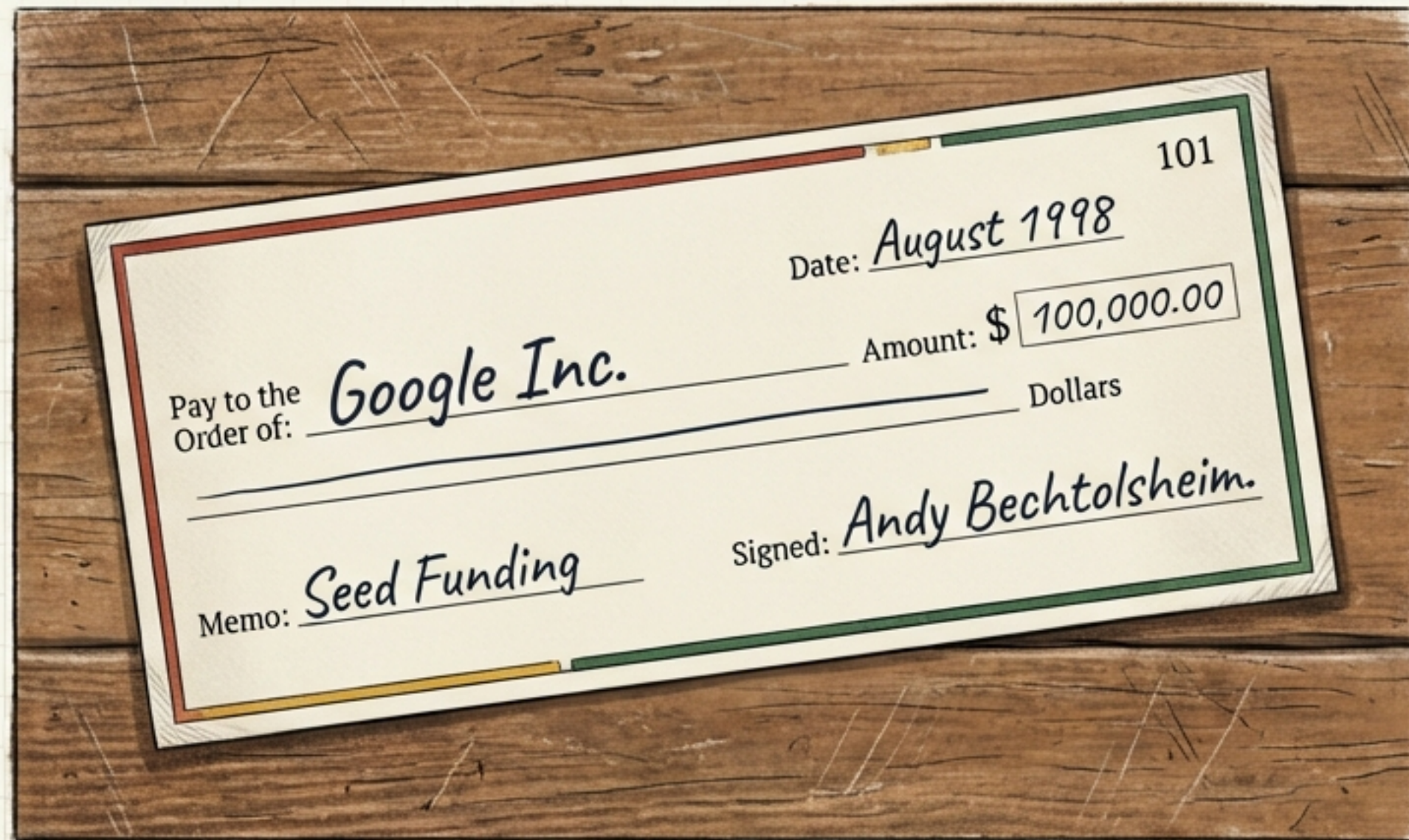
Name Origin: 'Googol' (10 to the power of 100). Misspelled as 'Google' during domain registration.



The Design: Sparse white space because they didn't know HTML.

The "I'm Feeling Lucky" button was a brazen display of confidence.

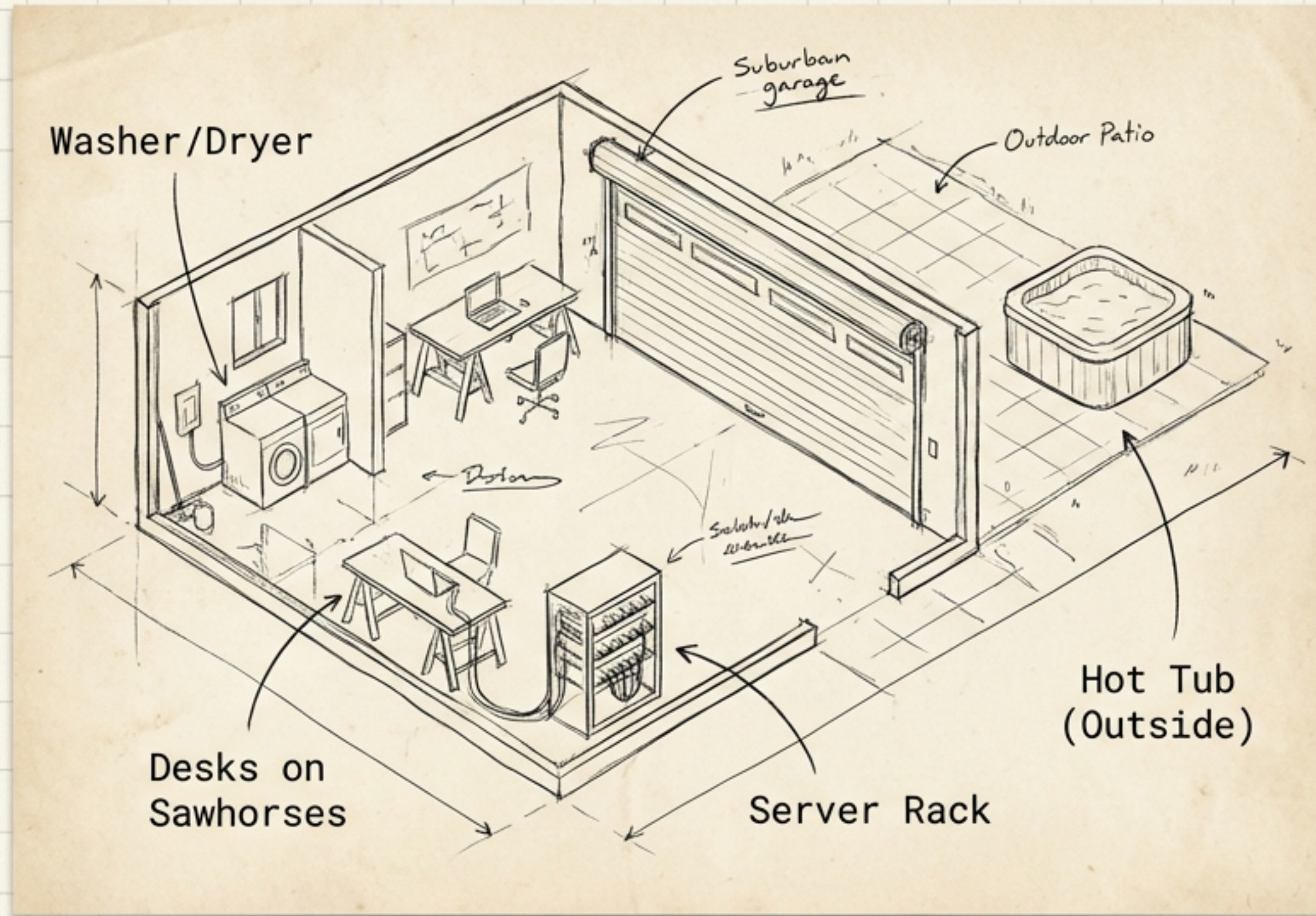
The \$100,000 Check.



The Moment: Sun Microsystems co-founder Andy Bechtolsheim saw a demo on a porch, was in a rush, and wrote the check immediately.

The Problem: “Google Inc.” didn’t exist yet. The check sat in Page’s dorm drawer for a month until they could incorporate.

232 Santa Margarita Street



Landlord: Susan Wojcicki
(Rent: \$1,700/month)

The Team: Larry, Sergey,
and Craig Silverstein
(Employee #1)

The Shift: The “Stanford
Research Project” officially
becomes “Google.com: The
Next Generation Internet
Search Company.”

Focus on the User.

Don't Be Evil.

Unlike competitors chasing “stickiness,” Google believed that if you solved the problem of finding information, the revenue would follow.

Key Takeaways.



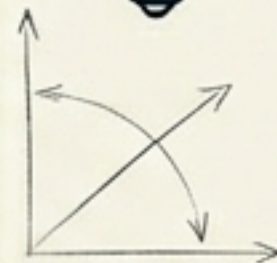
Innovation Requires Context:

PageRank worked because it treated the web as a graph of citations, not a library of text.



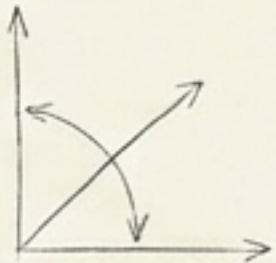
Commercialization is Vital:

Influenced by Tesla, Page knew that invention without business success is a tragedy.



Counter-Intuitive Wins:

Competitors wanted stickiness (bad search). Google wanted speed (good search). The user-centric model won.



Scarcity Breeds Creativity:

From Lego servers to garage offices, resource constraints drove their unique engineering culture.

